KEVIN KWAN

Software Engineer II | Conversational AI & Voice Automation
kevinkwan@gatech.edu | Atlanta, GA | kevin-kwan.github.io | linkedin.com/in/kevinkwan3 | github.com/Kevin-Kwan

SUMMARY

Software Engineer II at GEICO — conversational AI and voice automation. Build and operate the Sierra AI billing voice agent and Amazon Connect IVR on AWS: 436,000+ conversations at ∼51% containment, ∼30,000 calls/week.

WORK EXPERIENCE

GEICO July 2024 – Present
Atlanta, GA (Remote)
Software Engineer II — Conversational AI & IVR February 2026 – Present
Software Engineer I — Technology Development Program July 2024 – February 2026

- Operate the Sierra AI billing voice agent (live since May 2026) as its largest contributor, handling ∼30,000 GEICO billing calls per week — payments, rescheduling, AutoPay, billing questions — in English and Spanish; contained ∼51% of 436,000+ conversations without a live-agent transfer (May–Oct 2026), trending 48% → 52% as weekly volume grew ∼10×.
- Delivered end-to-end make-payment for the agent (saved method, pay-now or scheduled, debit authorization, declined-/expired-card recovery, payments on policies pending cancellation): 22,600+ payments at 92–95% authorization success; shipped net-new Pause AutoPay, 82,800 reschedules, and 192,000 info answers at 98% tool success.
- Re-architected the billing agent from regex-/classifier-based routing to fact-based journeys (178 files, ∼20,000 net lines removed) and built simulation-based regression suites as release quality gates, plus automated code-review follow-up, keeping high-velocity releases safe.
- Fixed 86 of 91 production and UT1 billing bugs at the root cause as the billing agent's largest contributor; reviewed code for 12+ engineers and onboarded two engineers onto the Sierra platform.
- Drove production reliability as on-call owner — release validation, live IVR testing, outage response and rollbacks, and postmortems; led a conversation-experience review across the agent's call volume that ranked and closed its biggest customer-experience gaps.
- Migrated GEICO's legacy Cisco UCCE and Nuance IVR to Amazon Connect on AWS, engineering Java traffic-redirection logic for gradual live caller cutover alongside legacy endpoint and Amazon Lex pattern migrations.
- Led the Proof of Coverage IVR (∼80% containment) and built the AWS self-service payment IVR (∼60% containment), owning the Proof of Coverage disaster-recovery rebuild pipeline and delivering a zero-amount-due flow, automated declined-payment retry flows, and new card tokenization.
- Engineered high-throughput, asynchronous voice-workflow orchestration using AWS Lambda, API Gateway, DynamoDB, and Amazon Lex (JavaScript code hooks) across Amazon Connect contact flows, securing customer PII and payment APIs with AES-256 encryption and automating infrastructure deployments via Terraform.

Publicis Sapient (Publicis Groupe) June 2023 – August 2023
Software Engineer Intern | Atlanta, GA

- Engineered a WhatsApp AI chatbot for the Travel & Hospitality industry using Google Vertex AI's PaLM LLM, LangChain, MongoDB, and external APIs to deliver live data and recommendations for Atlanta attractions, hotels, and restaurants, achieving 91% response accuracy — outperforming the team in Boston, as well as all the other teams in the Atlanta office.
- Architected Python/Flask RESTful APIs bridging MongoDB, the AI chatbot backend, and an interactive map website, automating CI/CD pipelines via GitLab and Docker to deploy on Google App Engine and Firebase Hosting.

Georgia Tech EcoCAR January 2022 – May 2023
Software Engineer, Connected & Automated Vehicles (V2X Subteam) | EcoCAR (Electric Vehicle) Mobility Challenge | Atlanta, GA

- Engineered a camera/sensor state machine enabling automated stop/go decisions across traffic-light scenarios, improving fuel efficiency and ride experience, helping Georgia Tech secure its 1st-place finish in the 2022 EcoCAR Mobility Challenge — a $1M program sponsored by the U.S. Department of Energy.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C#, C++
Conversational AI & Voice: Sierra AI, Amazon Connect, Amazon Lex, LangChain, Google Vertex AI (PaLM), OpenAI, Cyara
AWS & Cloud Infrastructure: AWS (Lambda, DynamoDB, API Gateway, S3, CloudWatch), Google Cloud Platform (App Engine, Firebase), Microsoft Azure, Cloudflare
Web, APIs & Data: REST APIs, Flask, Node.js, React, Next.js, Tailwind CSS, MySQL, MongoDB, Cloud Firestore
DevOps & Tooling: Terraform, Docker, CI/CD, Git, GitHub, GitLab, SonarQube, Cursor, Claude Code

EDUCATION

Georgia Institute of Technology — Atlanta, GA August 2021 – May 2024

- Bachelor of Science in Computer Science, College of Computing; Threads: Intelligence and Media; *summa cum laude* GPA: 3.9
- Zell Miller Scholarship, Graduated with Highest Honors, Dean's List, Faculty Honors